



Open Data, Design, & Development at the Office of Natural Resources Revenue

Reflections from our Virtual Student Federal Service (VSFS) user experience intern

May 14, 2020

Hello everyone! I am, Alena Huang, a student at the University of Notre Dame studying Visual Communication Design. I joined the team as a User Experience Intern for 9 months during the school year.

What I didn't know when I started

Coming into the internship, I didn't have too much experience in user experience, but it was something I was interested in. I had some experience in HTML/CSS and web design, but other than that, I had a lot to learn about UX!

Projects I've worked on

I've learned about all the different aspects of user experience and how they all fit within the process of design.

Content audit and sitemap

The very first thing that I worked on was a content audit and [sitemap](#) for ONRR.gov. This was really helpful in understanding the content and getting a sense of the scale of the information that the ONRR website provided. It was so surprising to see how many files were available to the public!

Snapshots of the content audit:

Homepage (onrr.gov)	0.0	
About	1.0	
	ONRR Organizational Chart	1.0.1
	ONRR FY 2017 Budget Justification	1.0.2
	ONRR Highlights	1.0.3
	Overview (about ONRR)	1.0.4
	Committees & Coordination	1.0.5
	Employment Opportunities	1.0.6
	Freedom of Information	1.0.7
	Leadership	1.0.8
	Natural Resource Revenue	
	Office Locations	1.0.9
	Presentations & Testing	1.0.10
	Press Release	1.0.11
	Rules & Regulation	1.0.12
Valuation	2.0	
	Indian Gas Index Zone	2.0.1
	Indian Gas Major Portion	2.0.2
	Indian Oil Major Portion	2.0.3
	Federal Gas Index Option	2.0.4
	NYMEX Oil Prices	2.0.5
	Unbundling	2.0.6

id	URL	Title	File Type	Accessibility	Description	Word Count	Reading Level	Notes (if any)
2.0.1	https://www.onrr.gov/Valuation	Indian Index Zones Natural Gas Price	html		shows monthly natural gas prices for six Indian Index Zones	40	11	index zones
2.0.1.1	https://www.onrr.gov/Law	Federal Register on June 23, 2011	pdf					says its
2.0.1.2	http://www.ecfr.gov/cgi-bin	30 CFR § 1206.172	-					
2.0.2	https://www.onrr.gov/Valuation	Indian Major Portion Gas Prices (Indian	html		shows a table with major portion prices and due dates	15	10	took me
2.0.3	https://www.onrr.gov/Valuation	Indian Oil Index Based Major Portion	html		shows a table with designated area and products, pop up of	31	7	should be
2.0.3.1	https://www.onrr.gov/Valuation	Reporter Training Presentation	pdf					
2.0.3.2	https://www.onrr.gov/Valuation	Map of all designated areas	pdf					
2.0.3.3	https://www.onrr.gov/PDF	Dear Reporter Letter - Indian Oil Valuation	pdf	not accessible				
2.0.4	https://www.onrr.gov/Valuation	Federal Gas Index Option	html		description of exclusions for natural gas index and approved gas	165	8	keep co
2.0.5	https://www.onrr.gov/Valuation	NYMEX Oil Prices	html		interactive graph with data 2004-2019 and downloadable excel	44	9	chart to
2.0.5.1	https://www.onrr.gov/Valuation	Crude Oil Market Centers	html		table of approved crude oil market centers	58	9	
2.0.5.2	https://www.ecfr.gov/cgi-bin	30 CFR § 1206.113	-					
2.0.6	https://www.onrr.gov/unbundling	Unbundling (Unbundling Home)	html		pop up with limitations (299 words, 14 grade level), filterable	101	8	
2.0.6.1	https://www.onrr.gov/unbundling	Standardized UCA page	html		describes what a standardized UCA and lessees are, with a table	218	11	links for
2.0.6.2	https://www.onrr.gov/unbundling	Methodology & Reporter Letters	html		page with documents of calculating UCA and reporter letters	50	8	
2.0.6.2.1	https://www.onrr.gov/unbundling	How to Calculate a Transportation UCA	pdf					
2.0.6.2.2	https://www.onrr.gov/unbundling	How to Calculate a Processing UCA	pdf					
2.0.6.2.3	https://www.onrr.gov/unbundling	List of Engineering Data Needs	pdf					
2.0.6.2.4	https://www.onrr.gov/unbundling	List of Accounting and Cost Data Needs	pdf					
2.0.6.2.5	https://www.onrr.gov/PDF	Royalty on Gas Used or Lost Along a	pdf	not accessible				
2.0.6.2.6	https://www.onrr.gov/unbundling	Guidance on Valuing Gas for Royalty	pdf	not accessible				
2.0.6.2.7	https://www.onrr.gov/unbundling	Guidance on Valuing Gas for Royalty	pdf	not accessible				
2.0.6.3	https://www.onrr.gov/unbundling	E-Blasts and Updates	html		list of e-blasts from 2013-2018 with search bar	1000	9	
2.0.6.4	https://www.onrr.gov/unbundling	Willow Creek Gas Plant	pdf					
2.0.6.5	https://www.onrr.gov/unbundling	Chipeta Plant	pdf					
2.0.6.6	https://www.onrr.gov/unbundling	Blacks Fork	pdf					
2.0.6.7	https://www.onrr.gov/unbundling	San Juan Conventional Transportation	pdf					
2.0.6.8	https://www.onrr.gov/unbundling	Echo Springs Gas Plant	pdf					

User research

Throughout the internship, I learned how not only to create questions for interviews, but also how to conduct user interviews and testing. I enjoyed user interviews because this is where you get a sense of how users interact with the product. After researching through user interviews, analysis reports (reports from old surveys, comparative analysis reports, etc.), and site analytics, my team and I were able to come up with [five personas](#) and their [user flows](#) for ONRR.gov. Additionally, I recently created a card sort to better understand how users grouped information on the website!

One of the personas that we created, including pain points and main tasks:

User Type 1: Operator (Reporter)



Reporters are people who work for companies, typically split into three categories: production, royalty and solids. They generally come to the site to file reports, as well as review how to report through the handbook and training information. Reporters may also have questions and they look for specific contacts to answer questions on reports.

Goals

- Files OGOR and 2014 reports (EMARF, Form ONRR 4444, W-9, etc)
- Enter adjustments to account for incoming inventory
- Find specific contacts to answer further questions on reporting
- Review handbook and Training information

Pain Points

- Finding who to contact, with the most up-to-date contact information
- Finding the correct reporting forms
- Finding the correct code numbers

Scenarios

- The reporter has a 2014 report that is due next Friday and needs to file, so they go to onrr.gov to file the report. They have a question about the report so they look for the contact information of the Royalty-specific worker at ONRR to call them.
- The reporter needs to enter adjustments to the account for incoming inventory. They log onto the ONRR ecommerce portal to make these changes. The reporter follows the Reporter Checklist and reporting documents to review how to report correctly.

Visual design & interaction design

Earlier in my internship, I helped in brainstorming different ways to visualize data on the [Natural Resource Revenue Data website](#), which is the data website that the ONRR team manages. Under the Explore data page, I helped design concepts for cards that will allow users to easily compare data between different states and regions. Near the end of my internship, I was able to redesign the [style guide](#) of the ONRR website, including colors, fonts,

and other UI elements. After creating the guide, I created prototypes for the [homepage](#), [contact page](#), and [reporter checklist page](#) that showcase basic visual design and information architecture.

Style tile:

ONRR Style Guide



Font, Source Sans Pro

Site Header

Font, Vollkorn

Site Subheader

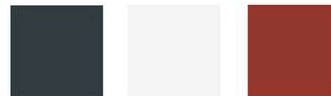
Font, Source Sans Pro

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut aliquip ex ea commodo consequat. Duis autem vel eum iriure dolor in hendrerit in vulputate velit esse molestie consequat, vel illum dolore.

Primary Colors



Background Colors



Alert Color (used rarely)

Lines



Buttons and Links



[This is an example of a link](#)

Favorite part

Although my internship was completely virtual, I enjoyed getting to know everyone on the ONRR team. Everyone is so welcoming and willing to teach me more about the process of developing a live, open-source government product. Whenever I went onto their conference calls for sync ups, they always included me in the conversation. I would especially like to thank Shannon McHarg and Lindsay Goldstein for being amazing mentors, as well as Annie Persson for being the best co-intern! I am incredibly thankful for everything that they have taught me and for giving me the opportunity to really dive deep into the process of user experience.

What's next

UX/UI Design is definitely on my radar; I am excited to take what I learned from this internship and use my skills in future projects!

Note : Reference in this blog to any specific commercial product, process, or service, is for the information and convenience of the public, and does not constitute endorsement, recommendation, or favoring by the Department of the Interior.



Alena Huang: User Experience Intern at the Office of Natural Resources Revenue.

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[Re-evaluating how we work: bringing the whole team along >](#)

[Home](#)



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From the team that works on Natural Resources Revenue Data and ONRR.gov. Have feedback or a blog post idea? Drop us a line.

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